

YUGUDA MUHAMMED

AI Engineer · Production LLM Systems · ML Infrastructure & Python

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PROFESSIONAL SUMMARY

AI Engineer with 5+ years building and deploying LLM-powered systems

TECHNICAL SKILLS

Languages: Python (5+ yrs) modelling, data pipelines, automation, API development; SQL

LLMs & ML: LLM application & agent development (LangChain, LangGraph, LangFlow, CrewAI); RAG, embeddings & semantic retrieval; prompt optimisation, model evaluation & output validation; open-weight models (Gemma); Hugging Face Transformers, scikit-learn, NumPy, Pandas

Inference & Serving: FastAPI / Django REST / Flask inference APIs, JSON contracts & OpenAPI docs; async task execution (Celery); Redis caching; low-latency serving under concurrent load

Vector & Data: Qdrant, ChromaDB, Supabase pgvector; PostgreSQL (schema design, query optimisation), Redis, MySQL; end-to-end pipelines over large structured & unstructured datasets

MLOps & Deployment: Docker, Kubernetes, GitHub Actions CI/CD, AWS (EC2, S3, RDS, ElastiCache), GCP; zero-downtime deployment, automated testing

Monitoring & Governance: Latency / performance monitoring, model & dataset versioning, evaluation benchmarks, schema compliance & output validation, technical & architecture documentation

Collaboration: Git & code review, Agile / Kanban, cross-functional distributed remote teams

PROFESSIONAL EXPERIENCE

Lead AI Engineer

Jul 2024 – Present

Adept Engineering Solutions, USA · Remote

- Lead design and delivery of an end-to-end, production LLM system that ingests complex solicitation documents, applies multi-step reasoning agents over structured (JSON) data, and returns scored, validated outputs via FastAPI on AWS.
- Drove output quality from 30% to 90%+** through iterative model evaluation, prompt optimisation, and structured output validation establishing repeatable evaluation benchmarks for every release.
- Built ingestion, preprocessing, embedding, and semantic-retrieval pipelines over noisy real-world data, with robust handling of edge cases and failure modes.
- Defined JSON API contracts and data schemas for all model inputs/outputs; authored system-architecture specs and methodology docs used across engineering teams.
- Engineered for production load (~100 concurrent users): Redis caching, horizontal scaling, and latency monitoring to maintain low-latency responses.
- Own delivery milestones across a distributed remote team and communicate technical decisions to engineering and non-technical stakeholders.

AI / Backend Engineer

May 2022 – Jan 2025

Bitsaac Inc. · Remote · FinTech / E-commerce

- Built RAG-powered analysis pipelines semantic retrieval systems using LLMs as a reasoning layer over structured data.
- Improved vector retrieval performance by 30%** via embedding-model evaluation, dataset-quality work, and ChromaDB / Qdrant tuning.
- Shipped RESTful JSON APIs for model operationalisation; integrated external AI providers with robust error handling and fallback behaviour to keep services reliable under failure.

- Maintained comprehensive test coverage and methodology documentation; collaborated across engineering, product, and data in a fully remote Agile environment.

LLM Agent Trainer & Evaluator

Sept 2024 – Oct 2024

Highbrow Technology · Contract · Remote

- Defined evaluation criteria and quality-assured structured LLM outputs for correctness, reasoning quality, and schema compliance; documented findings in clear written reports directly applicable to ML evaluation and governance.
- Applied prompt engineering and data refinement to improve model reasoning accuracy; collaborated asynchronously using Git for version control.

SELECTED PROJECTS

Government Proposal Generation Production LLM Reasoning System

Python · FastAPI · LangChain / LangGraph · OpenAI · PostgreSQL · Redis · Docker · AWS

- Autonomous multi-step LLM agent that ingests raw PDF/JSON documents, structures unstructured data, reasons over it, and produces scored, formatted outputs through a production REST API.
- Designed JSON schemas for every I/O stage; deployed on AWS with Docker, monitoring, performance benchmarking, and iterative optimisation cycles with measurable quality outcomes.

DevMemory Self-Hostable AI Infrastructure (MCP Server) (<https://github.com/Yuguda999/devmemory>)

Python · Model Context Protocol · FastAPI · SQLAlchemy · PostgreSQL / SQLite · Docker · GitHub Actions

- Built a **self-hostable, multi-user memory server that runs fully locally with no external API dependency**, exposing reusable tooling consumed across multiple AI clients (Claude, Cursor, Windsurf, and others).
- Implemented JWT / API-key auth, tiered quota enforcement, a monitoring web dashboard, Alembic migrations, and CI a reusable internal-infrastructure pattern that cuts external-API reliance and time-to-deploy for new AI tooling.
- Published to PyPI with cross-platform installers and a full REST API (pypi.org/project/devmemory).

Eeva Multimodal WhatsApp AI Assistant (<https://github.com/Yuguda999/eeva>)

Python · FastAPI · Gemma-4 (open-weight vision model) · SQLite · Meta Cloud API

- Multimodal agent that analyses product/business images and generates marketing content over WhatsApp, driven by a stateful message router / state machine for iterative refinement.
- Built on an **open-weight model (Gemma-4)** rather than a proprietary API demonstrating ability to run AI workflows on self-hostable / open models.

CampusPQ Multimodal AI Learning Platform

Python · FastAPI · OpenAI · PostgreSQL · VectorDB · AWS · CI/CD · campuspq.com

- Production multimodal pipelines (PDF / image / audio / video) with LLM-based analysis; **scaled to 5k + processed questions at a 98% success rate**.
- Built the operationalisation layer REST / JSON APIs, performance tracking, and automated testing.

EDUCATION

B.Sc. Physics Ahmadu Bello University, Zaria

2018 – 2024

Physics background underpins the quantitative rigour, systems thinking, and methodical documentation behind robust applied ML engineering.